SURJECTION IN THE FORMAL COHERENT CATEGORY IS TESTED ON THE FIRST INFINITESIMAL LAYER

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9614-9621 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 9623-9628, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Feeds the algebraization suite (L16158 collected, L16161 collected); sequenced after its input lemma inverse-systems-abelian which is already on the shelf (wave 10).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma inverse-systems-abelian (levelwise etale-local abelian structure of the formal category); Cohomology of Schemes, lemma inverse-systems-surjective (the schemes Nakayama test).

Source reference. Original lemma 087Y: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9628–9628. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 089P: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9563–9573. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S and let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals. A map $(\mathcal{F}_n) \to (\mathcal{G}_n)$ is surjective in $Coh(X, \mathcal{I})$ if and only if $\mathcal{F}_1 \to \mathcal{G}_1$ is surjective.*

Proof. We can check on an affine étale cover of X by Lemma 089P. Thus we reduce to the case of schemes which is Cohomology of Schemes, Lemma 087Y. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.